Stochastic Processes I:
Random Walks & Markov Chains

Adrià Garcés

Física de la Matèria Condensada, Universitat de Barcelona
University of Barcelona *Institute of Complex Systems* (UBICS)

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Motivation, Definitions & Random Walks

1.1 Motivation

"Stochastic" is a partial synonim of "anything" in nature: physics, biology, medicine, sociology, finance ... Mostly evertyhing.

- The art of conjecturing, aim and guess. First appears with Jakob Bernoulli (1713) and some martingale strategies used for betting (France, XVIII century).
- Botanist R. Brown looking at pollen immersed in water (video)
- Bachelier first models asset prices with what later on would be devloped to become Brownian motion (video)
- Cell division, tumor growth (video)
- Population (birth, death, atomic species), ecological richness, resource abundance.

Stochastic is the language of the probabilistic, involving randomness. Not deterministic, but provided with a measure of likelyhood.

1.2 Definitions

Random Variables (RV)

- The RV X is said to be *discrete* when its sample space Ω_X is finite or countably infinite.
- The RV X is said to be continuous when its sample space Ω_X is uncountably infinite (interval).

Stochastic Process

Vaguely speaking, *a stochastic process is a collection of (time) indexed RVs, i.e.* $\{X_t\}_{t \in T}$.

Families (nature of RV and time indexing)

RV	Discrete Time	Continuous Time
Discrete	Random Walks (RW), DTMC	CTMC, Birth & Death
Continuous	AR(p), ARCH(q)	(G)BM, Heston Model

1.2 Definitions – Formal

Formally speaking, *a stochastic process is a collection of (time) indexed RVs, i.e.* $\{X_t\}_{t \in T}$, **but**

Definition (Reminder)

Suppose that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, and $X : \Omega \to \mathbb{R}$ is a RV. Recall that this means that Ω is a space, $\mathcal{F}$ is a $\sigma-$algebra of subsets of Ω, $\mathbb{P}$ is a countably additive, non–negative measure on $(\Omega, \mathcal{F})$ with total mass $\mathbb{P}(\Omega) = 1$, and X is a measurable function, i.e., $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$ for every Borel set $B \in \mathcal{B}(\mathbb{R})$.

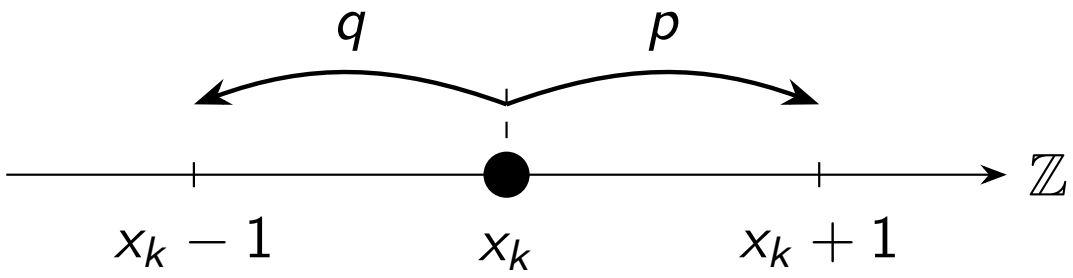
Definition (Stochastic process)

Suppose that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and where $T \in \mathbb{R}$ may have finite or infinite cardinality. Suppose further that for each $t \in T$, there is a RV $X_t : \Omega \to \mathbb{R}$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$. The function $X : T \times \Omega \to \mathbb{R}$ defined by $X(t, \omega) = X_t(\omega)$ is called a stochastic process with indexing set T, and is written $X = \{X_t\}_{t \in T}$.

1.3 Random Walks (RW) – Introduction

The (standard) Random Walk (RW) is an elementary example of (discrete time) discrete stochastic process.

- A *"walker"* starts at a given position on the $\mathbb{Z}$ line, e.g. X_0 at time 0. At each step of time (unit) it moves to the right $(+1)$ or left (-1) with probabilities p and q respectively.
- At each step, $X_{k+1} = X_k + Z_{k+1}$, with $p(Z_{k+1} = 1) = p$, $p(Z_{k+1} = -1) = q = 1 - p$ (i.i.d).



- The final position of the walker after n steps is

$$X_n = X_0 + S_n, \quad S_n = \sum_{\ell=1}^{n} Z_\ell \in \{-n, -n+1, \ldots, n-1, n\} \qquad (1)$$

1.3 Random Walks (RW) – MGF, Mean & Variance

The moment generating function (MGF) of a random variable X provides its full statistical description (the moments).

$$M(u) \equiv \mathbb{E}[e^{uX}], \quad \mathbb{E}[X^k] = \frac{d^k M(u)}{du^k}\Big|_{u=0} \tag{2}$$

For a RW $X_n = X_0 + \sum_{\ell=1}^{n} Z_\ell$, the MGF is (remember, Z_ℓ i.i.d, set $X_0 = 0$ without loss of generality),

$$M_n(u) = [pe^u + qe^{-u}]^n, \quad \mathbb{E}[X_n^k] = \sum_{\ell=0}^{n} \binom{n}{\ell} p^\ell q^{n-\ell} (2\ell - n)^k \tag{3}$$

Hence, the RW along the integer line has mean and variance

$$\mathbb{E}[X_n] = n(2p-1) \tag{4}$$

$$\mathbb{V}[X_n] = \mathbb{E}[X_n^2] - \mathbb{E}[X_n]^2 = 4np(1-p) \text{ (MSD)}. \tag{5}$$

Remaks: Non–biased walker? Extremely biased walker? **Bonus:** the cumulant generating function (CGF) $K(u) \equiv \log M(u)$ and its derivatives provide information about the cumulants, κ_k ($k = 2$ is the variance, etc.).

1.3 Random Walks (RW) – Binomial map

The RW can be mapped into the Binomial model:

- The variable final position of the walker after n steps can be expressed in terms of steps to the right N_+, and steps to the left N_-

$$X_n = \sum_{\ell=1}^{n} Z_\ell = +1+1+(-1)+1+(-1)+\ldots+1$$
$$\equiv +1N_+ + (-1)N_- = N_+ - N_-, \tag{6}$$

  where $N_+ + N_- = n$, i.e. $X_n = 2N_+ - n$.

- N_+ is binomially distributed, since it represents number of successes after n trials with probability p, $N_+ \sim B(n, p)$.
- As a consequence, the probability of finding the walker at position k after n steps is

$$p(X_n = k) = p\left(N_+ = \frac{n+k}{2}\right) = \binom{n}{\frac{n+k}{2}} p^{\frac{n+k}{2}} q^{\frac{n-k}{2}}. \tag{7}$$

1.3 Random Walks (RW) – Hitting probabilities

The Gambler's ruin. Canonical model of a hitting probability problem.

> Consider the familiar gambler who wins or loses a dollar with probabilities p and q, respectively. Let his initial capital be z and let him play against an adversary with initial capital $a - z$, so that the combined capital is a. The game continues until the gambler's capital either is reduced to zero or has increased to a, that is, until one of the two players is ruined. We are interested in the probability of the gambler's ruin and the probability distribution of the duration of the game. This is *the classical ruin problem.*

Figure: The Gabler's ruin problem, extract from Feller's "*An Introduction to Probability Theory, V1*".

- Identical to the problem of a RW stopping at either $x = 0$ or $x = a$.
- We shall use Feller's language, although the notation of RW, i.e. X as the position of the RW and so on.

1.3 Random Walks (RW) – Hitting probabilities

Let x_0 denote the initial position (or gambler's initial captial) and let Pr_{0,x_0} and Pr_{a,x_0} be hitting $x = 0$ (ruin) and $x = a$ (win) at the end of the game if the RW's initial position is $x_0 \in \{1, \ldots, a-1\}$. Then,

$$\mathrm{Pr}_{0,x_0} + \mathrm{Pr}_{a,x_0} = 1 \tag{8}$$

$$\mathrm{Pr}_{0,x_0} = q\mathrm{Pr}_{0,x_0-1} + p\mathrm{Pr}_{0,x_0+1}, \quad \mathrm{Pr}_{0,0} = 1, \mathrm{Pr}_{0,a} = 0, \tag{9}$$

Solution

The general solution complying with the boundary conditions $\mathrm{Pr}_{0,0} = 1$, $\mathrm{Pr}_{0,a} = 0$ is

$$\mathrm{Pr}_{0,x_0} = \frac{(q/p)^a - (q/p)^{x_0}}{(q/p)^a - 1} \tag{10}$$

- *Exercise:* find the solution of the symmetric case $p = q = \frac{1}{2}$.
- Strongly related to other interesting stochastic variables such as the first hitting (passage) times (duration of game).

1.3 Random Walks (RW) – Scaling

Scaling: behaviour of a model or theory wheen smoothly changing the scale of observation, i.e. $x \to \varepsilon_x x$, $t \to \varepsilon_t t$ (x, t already have dimension or characteristic scale).

Providing a scale to RW

The simple RW model does not include time or lengthscales (measured both in steps). Consider the symmetric RW and define characteristic space and time scales a and τ, and $\tilde{X} = aX$, $t = \tau n$. Then,

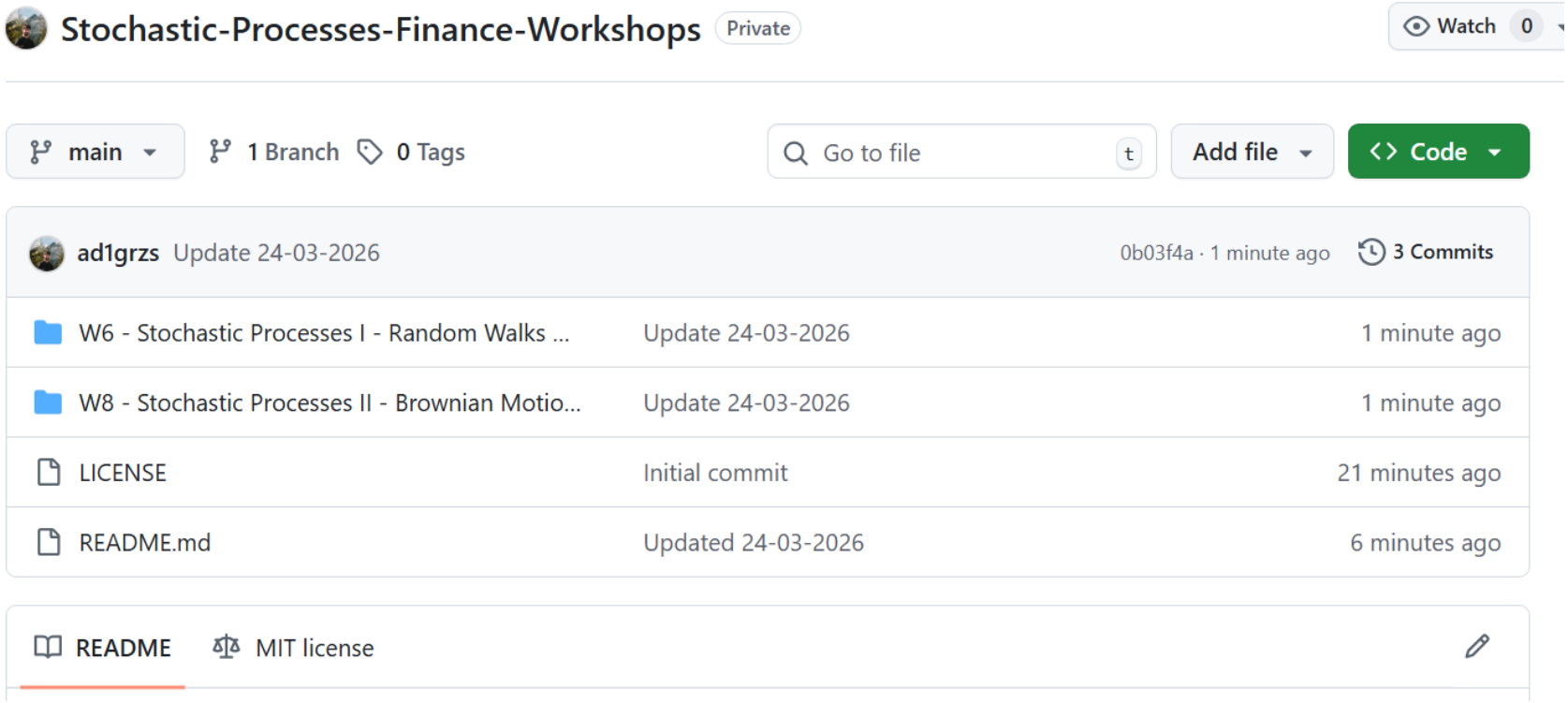
$$\mathbb{E}[\tilde{X}_t] = 0, \quad \mathbb{V}[\tilde{X}_t] = \left(\frac{a^2}{\tau}\right) t. \tag{11}$$

Scale transformations

We define the scale transformed variables $\tilde{X}_{\varepsilon_x} = \varepsilon_x \tilde{X}$ and $t_{\varepsilon_t} = \varepsilon_t t$. The variance $\mathbb{V}[\tilde{X}_{\varepsilon_x, t_{\varepsilon_t}}] = (\varepsilon_x^2/\varepsilon_t)(a^2/\tau) t_{\varepsilon_t}$, meaning the RW is scale–invariant only and only if $\varepsilon_t = \varepsilon_x^2$.

1.4 Implementation of RWs using Python

Access the GitHub repo here.



1 - Closing

Multifractal random walk

E. Bacry
Centre de Mathématiques Appliquées, Ecole Polytechnique, 91128 Palaiseau Cedex, France

J. Delour and J. F. Muzy
Centre de Recherche Paul Pascal, Avenue Schweitzer, 33600 Pessac, France
(Received 22 May 2000; revised manuscript received 4 April 2001; published 17 July 2001)

We introduce a class of multifractal processes, referred to as multifractal random walks (MRWs). To our knowledge, it is the first multifractal process with continuous dilation invariance properties and stationary increments. MRWs are very attractive alternative processes to classical cascadelike multifractal models since they do not involve any particular scale ratio. The MRWs are indexed by four parameters that are shown to control in a very direct way the multifractal spectrum and the correlation structure of the increments. We briefly explain how, in the same way, one can build stationary multifractal processes or positive rand

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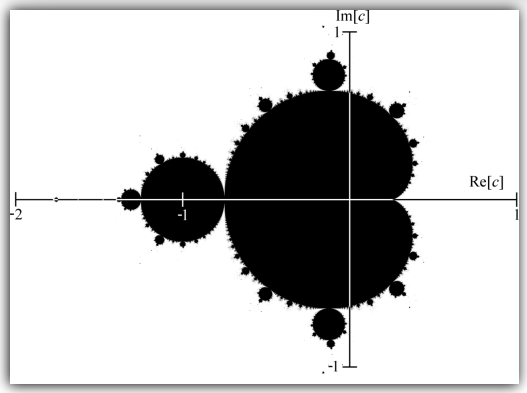
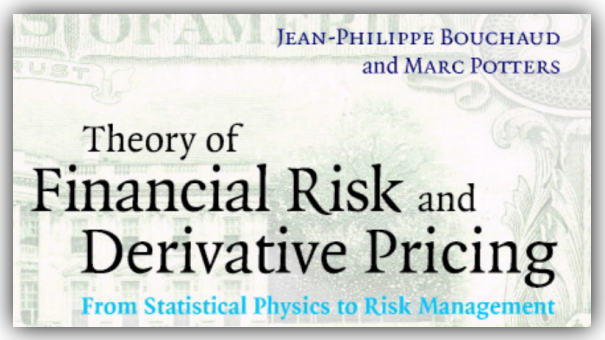


$$N\varepsilon^d = 1$$



7.4 Summary

- In financial markets, the volatility is not constant in time, but has random, log-normal like, fluctuations. These fluctuations cannot be described by a single time scale. Rather, their temporal correlation function is well fitted by a power-law. This power-law reflects the multi-scale nature of these fluctuations: high volatility periods can persist from a few hours to several months.
- From a mathematical point of view, the sum of increments with a finite stochastic volatility still converge to a Gaussian variable, although the kurtosis decays much more slowly than in the case of iid variables when the volatility has long range correlations. Accordingly, the empirical moments of price variations approach the Gaussian limit only very slowly with the time horizon, and their long time value is dominated by volatility fluctuations.
- The slow decay of the volatility correlation function, together with the observed approximate log-normal distribution of the volatility, leads to a 'multifractal-like' behaviour of the price variations, much as what is observed in turbulent hydro-dynamical flows.

Markovianity, Stationarity & Convergece

2.1 Definitions – Joint distribution and Markovianity

The time ordered stochastic process $\{X_t\}_{t\in T}$ is generally characterized (single realization) by the joint distribution function (take $T = \{0, \ldots, n\}$) and its marginals

$$p(X_0, X_1, \ldots, X_n), \quad p(X_k) = \sum_{\{X_m\}_{m\neq k}} p(X_0, X_1, \ldots, X_n). \tag{12}$$

Generally,

$$p(X_0, \ldots, X_n) = p(X_n|X_{n-1}, \ldots, X_0)p(X_{n-1}|X_{n-2}, \ldots, X_0) \ldots p(X_0). \tag{13}$$

Markovianity

A time ordered stochastic process is said to be Markovian if $p(X_m|X_{m-1}, \ldots, X_0) = p(X_m|X_{m-1})$.

$$p(X_0, \ldots, X_n) = p(X_0) \prod_{m=0}^{n-1} p(X_{m+1}|X_m). \tag{14}$$

2.1 Definitions – Markov Chains

The time evolution of a Markovian stochastic system is usually referred to as **Markovian Dynamics** or **Markov Chain**. Since the state at current time X_n only depends on the state of the immidiate past, X_{n-1}, the probability evolves according to,

$$p(X_n) = \sum_{\{X_{n-1}\}} p(X_n|X_{n-1})p(X_{n-1}). \tag{15}$$

When the possible states of the system are labeled with a given index $i = 1, \ldots, N$, i.e. $X_n = i \in \{1, \ldots, N\}$, we write

$$\boxed{p(X_n = i) = \sum_{j=1}^{N} p(X_n = i|X_{n-1} = j)p(X_{n-1} = j) \Leftrightarrow \mathbf{p}_n = W\mathbf{p}_{n-1},} \tag{16}$$

where $W \in \mathbb{R}^{N\times N}$ is called the transition rate matrix, with elements $W_{ij} = p(X_n = i|X_{n-1} = j)$ (transition rates, from state j to state i).

2.2 Stationarity & Convergence

If the transition matrix W does not depend on time (n), then it follows that

$$\mathbf{p}_n = W^n \mathbf{p}_0, \tag{17}$$

where $\mathbf{p}_0$ is the initial condition's probability distribution.

Stationarity

A system is said to have reached a *stationary state* if the probabilities remain invariant under time evolution, i.e. $\mathbf{p}_{n+1} = \mathbf{p}_n$. In order for this to happen there must exist $\mathbf{p}_{st}$ s.t. $\mathbf{p}_{st} = W\mathbf{p}_{st}$ (eigenvalue problem).

- Then $\mathbf{p}_{st}$ is the stationary distribution of the chain.
- The existence, uniqueness and positiveness of $\mathbf{p}_{st}$ is guaranted when $\sum_{i=1}^{N} W_{ij} = 1$ for all $j = 1, \ldots, N$ (irreducible, aperiodic).
- Inution; if for a Markov Chain $\{X_0, X_1, \ldots, X_n\}$, $X_0 \sim \mathbf{p}_{st}$, then for all n, $X_n \sim \mathbf{p}_{st}$.

2.2 Stationarity & Convergence

If the transition matrix W does not depend on time (n), then it follows that

$$\mathbf{p}_n = W^n \mathbf{p}_0, \tag{18}$$

where $\mathbf{p}_0$ is the initial condition's probability distribution.

Convergence

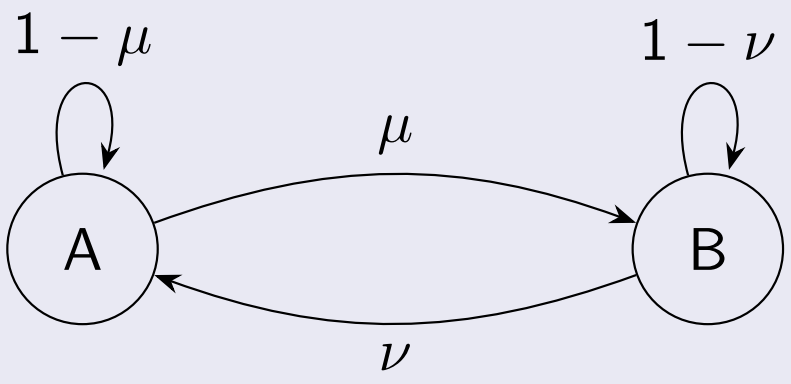
A powerful theorem claims any irreducible and aperiodic MC *converges* to its stationary distribution, i.e. $\lim_{n\to\infty} ||W^n \mathbf{p}_0 - \mathbf{p}_{st}||_{\text{tv}} = 0$. Handwaving;

- If W irreducible, aperiodic then W has eigenvalues $\lambda_N < \cdots < \lambda_2 < \lambda_1 = 1$. There exists $P = (\mathbf{p}_{st}, \mathbf{v}_2, \ldots, \mathbf{v}_N)$ s.t. $W = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \ldots, \lambda_N)$.
- Indeed

$$(\mathbf{p}_n)_\ell = (\mathbf{p}_{st})_\ell (P^{-1}\mathbf{p}_0)_1 + \sum_{m=2}^{N} \lambda_m^n (\mathbf{v}_m)_\ell (P^{-1}\mathbf{p}_0)_m \overset{n\to\infty}{\to} (\mathbf{p}_{st})_\ell. \tag{19}$$

Example exercise – *Two state MC*

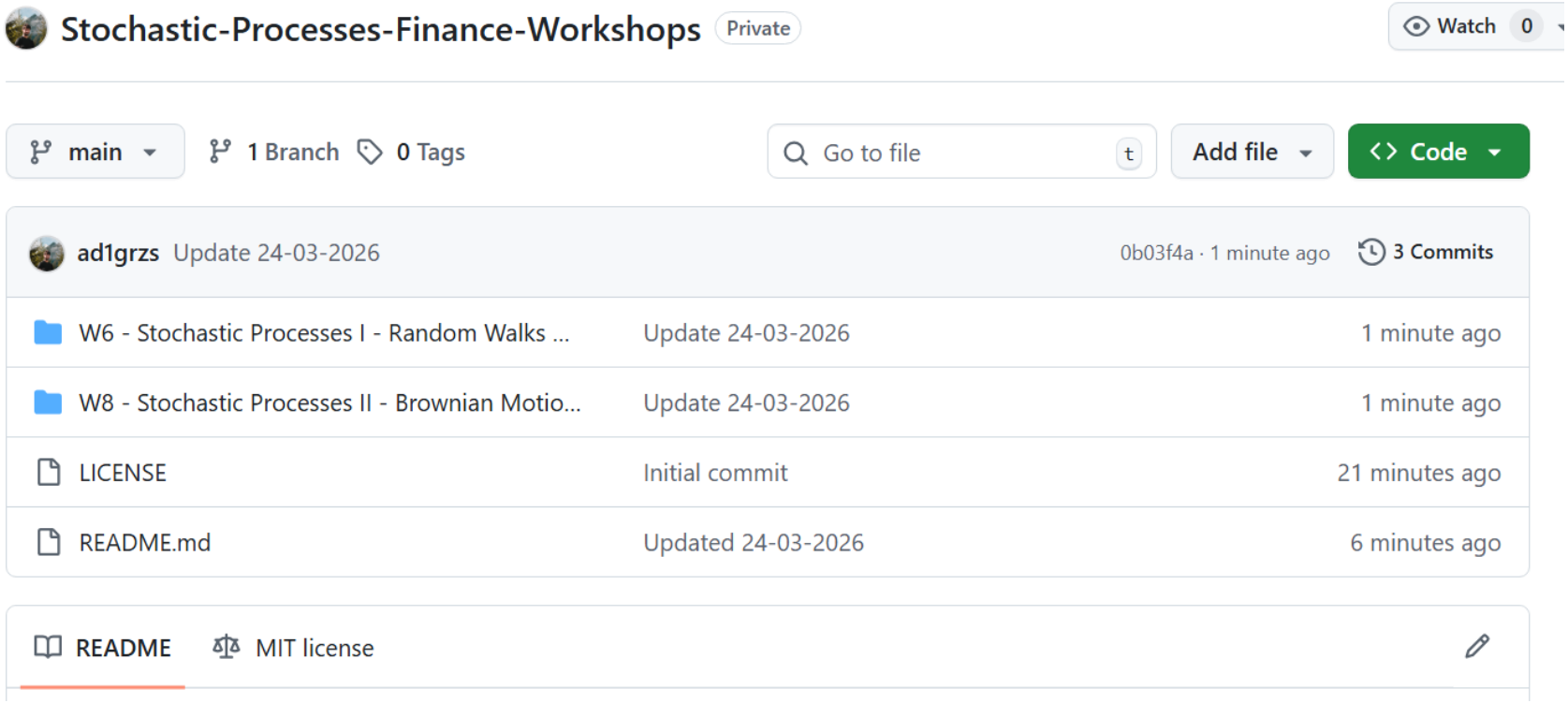
Pictorial description (common use in MC) of a two state MC, i.e. $X = A, B(1, 2)$.



- Shape of the transition matrix W?
- Show that $\mathbf{p}_{st} = \frac{1}{\mu+\nu}(\nu, \mu)^T$.
- Find the probability of finding $X = A$ at time n if the initial distribution is arbitrary.

2.3 Implementation of two state MCs in Python

Access the GitHub repo here.

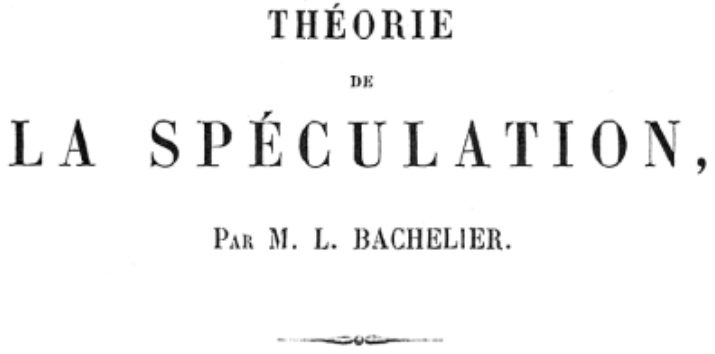


Financial Analogies

3.1 The beginning of mathematical finance

Louis Bachelier introduces a novel idea: the market can be described conceptually as a stochastic process.

- Asset prices S_t follow "diffusive" Brownian motion – the continuous analogous of a RW.
- Asset prices modelled $S_n = S_{n-1} + sZ_n$, with $Z_n = \pm 1$ with probabilities $p = q = \frac{1}{2}$ ($s \sim$ volatility).
- Bachelier's ideas didn't have immidiate interest, a continued tradition nor introduced new school of thought.


THÉORIE

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LA SPÉCULATION,

PAR M. L. BACHELIER.

3.1 Binomial asset evolution

An improvement guaranteing positiveness of asset prices S_t is the following (binomial) model,

$$S_n = S_{n-1} Z_n, \quad Z_n = \begin{cases} u > 1 & \text{probability } p \\ 0 < d < 1 & \text{probability } q = 1 - p \end{cases} \tag{20}$$

The final price after n steps is nothing but the product $S_n = S_0 \prod_{\ell=1}^{n} Z_\ell$, which at the same time can be written as

$$S_n = S_0 u^{N_+} d^{n-N_+}, \quad N_+ \sim \text{Binom}(n, p). \tag{21}$$

This has important consequences,

$$\boxed{\log\left(\frac{S_n}{S_0}\right) = N_+ \log u + (n - N+) \log d, \quad \log\left(\frac{S_n}{S_0}\right) \sim \text{Binom}(n, p)} \tag{22}$$

$\Rightarrow$ Lognormality as $n \gg 1$.

3.1 Binomial asset evolution

For a recombining tree with $d = u^{-1}$, the price of the asset after n steps can be written

$$S_n = S_0 u^{2N_+ - n} \equiv S_0 u^{X_n}, \quad \log \frac{S_n}{S_0} = X_n \log u. \tag{23}$$

where $X_n = 2N_+ - n$ is a simple Random Walk (sRW) with probabilities $p, q = 1 - p$. Full access to the moments and distribution.

$$\mathbb{E}\left[\left(\frac{S_n}{S_0}\right)^k\right] = \left(\frac{1 - p + u^{2k} p}{u^k}\right)^n. \tag{24}$$

A devastating consequence

Since,

$$\mathbb{E}\left[S_n / S_0\right] = u^{-n}[1 - p + u^2 p]^n \tag{25}$$

the line $p_c(u) = \frac{1}{1+u}$ separates a regime in which prices grow ($p > p_c(u)$, inflation) or drop ($p < p_c(u)$, deflation) on average.

3.1 Binomial asset evolution

For a recombining tree with $d = u^{-1}$, the price of the asset after n steps can be written

$$S_n = S_0 u^{2N_+ - n} \equiv S_0 u^{X_n}, \quad \log \frac{S_n}{S_0} = X_n \log u. \tag{26}$$

where $X_n = 2N_+ - n$ is a simple Random Walk (sRW) with probabilities $p, q = 1 - p$. Variance easier to understand in taking natural logarithm

$$\mathbb{E}\left[(\log S_n/S_0)^k\right] = \mathbb{E}[X_n^k](\log u)^k \tag{27}$$

Volatilities

Here,

$$\sigma_{\mathrm{B},n}^2 \equiv \mathbb{V}[\log(S_n/S_0)] = 4np(1-p)(\log u)^2. \tag{28}$$

At fixed u, volatility is maximum when $p = 1/2$, and takes value $\sigma_{\mathrm{B},n}^2 = n(\log u)^2$. Volatility grows monotonously with u for any fixed p ($u > 1$).

3.2 Mean reversion

Formal (finance) definition – Mean reversion: assumption that an asset's price converges to the to the average price over time. In Finance101 terms;

- If price is below the average in a given time window, it will go up (most probably).
- If price if above the average in a given time window, it will go down (most probably).

Conceptually speaking: this requires structural memory, fixed variances and the existence of stable equilibrium(s)

Examples

- Random Walks and Binomial evolution of prices **do not posses** mean reversion.
- The Ornstein Uhlenbeck process **possesses** mean reversion.

$$S_n = S_{n-1} + \kappa(\Theta - S_{n-1}) + sZ_n. \tag{29}$$

3.2 Martingales & Arbitrage

Formal definition – Martingale: a stochastic process $\{X_n\}$ is called a martingale if for any n, $\mathbb{E}[|X_n|] < \infty$ and

$$\mathbb{E}[X_{n+1}|\mathcal{F}_n] = X_n \quad \text{or for MC} \quad \mathbb{E}[X_{n+1}|X_n] = X_n. \tag{30}$$

Intutively speaking: the most probable value of X tomorrow will be the value it had today (same for prices, fair game – no arbitrage).

Examples

- The symmetric RW is a martingale, since $\mathbb{E}[X_{n+1}|X_n] = X_n$.
- The generally asymmetric RW is not a martingale, since $\mathbb{E}[X_{n+1}|X_n] = X_n + (2p - 1)$.
- The binomial asset evolution is a martingale when $d = 2 - u$ and when $p = q = 1/2$.
- (de Moivre's) The process $Y_n = (q/p)^{X_n}$ where X_n is an unfair coin toss with probabilities p (H) and $q = 1 - p$ (T) is a martingale with respect to the toss, $\mathbb{E}[Y_{n+1}|X_n] = Y_n$.